NIKOLA MILOSAVLJEVIĆ

+41 78 348 3224 | nikolamilosa20@gmail.com | GitHub (nikolamilosa) | linkedin Zürich, Switzerland

TECH STACK

Languages: Rust, Golang, Python, C# (.NET)
Databases: SQL (PostgreSQL, Mssql), NoSQL (MongoDB, Neo4j)
Automation: GitHub Actions, Ansible, Bazel, Airflow
IaaC: Terraform, Pulumi
Cloud: AWS, GCP, Azure, Kubernetes, Docker
Public work: dre, ic, decentralization

WORK EXPERIENCE

DFINITY Stiftung **Zurich, Switzerland**
Site Reliability Engineer ***Sep 2022 – Present***

- Designed and implemented a scalable service discovery module powering the company's **observability platform**, synchronizing global system state, integrating with **Prometheus** and **VictoriaMetrics**, and generating dynamic observability configurations.
- Architectured, built, and operated the company's end-to-end logging infrastructure; led migration from AWS to on-prem Kubernetes, reducing costs by **~$100K/month** and supporting **~10M prod logs/hour** and **~20M dev logs/hour**.
- Devised a local testing driver using QEMU and libvirt to spin up production-like environments, enabling 1:1 testing against production and shortening feedback loops while improving code quality.
- Constructed an automated release qualification pipeline, executing pre/post-upgrade performance tests in production-like environments to quantify regressions and guide release decisions.

DFINITY Stiftung
Backend Engineer

- Developed features reducing operational burden for node providers, including widely used direct node swaps and migrations that reduced key rotation time from days to seconds while improving security and flexibility.
- Introduced a linear optimization tool for decentralized network design, enabling constraint-based topology generation, trade-off modeling, and capacity planning for target network states.
- Engineered core components of the **company's cloud platform**, including a Web2 control panel for provisioning and managing sovereign cloud environments and a Web3 on-chain module for billing and **payments with stablecoin and asset swap integration**.
- Contributed to developer experience and internal tooling at the company, managing testnet infrastructure, improving build systems, building on-call utilities, and streamlining release and feature rollouts.

Sotex Solutions (outsourced to Schneider Electric Sevilla) **Novi Sad, Serbia**
Fullstack Software Engineer ***May 2021 – Sep 2022***

- Built a billing module for municipalities to generate utility bills across Kuwait, Sevilla, and the Netherlands, integrating IoT smart meters and handling **terabytes of data** with reliable, large-scale processing.
- Developed smart meter orchestration systems enabling users and technicians to remotely configure, monitor and diagnose devices using industry-standard protocols, improving observability and operation control at scale.

Sotex Box **Novi Sad, Serbia**
Fullstack Software Engineer (Founding Engineer) ***May 2024 – Sep 2025***

- Built and scaled a low-bandwidth ad delivery platform for constrained networks, running on minimal infrastructure (~$80/month) while supporting **~10K concurrent devices** and generating **~$15K/month** in revenue.

University of Novi Sad **Novi Sad, Serbia**
Teaching Assistant ***Sep 2023 – Sep 2024***

- Supported courses on **microservice architecture**, leading labs and assisting lectures for 300+ students across multiple university groups.
- Delivered introductory **cybersecurity** lectures to 150+ students, teaching core concepts across several university groups.

EDUCATION

Faculty of Technical Sciences **Novi Sad, Serbia**
Bachelor's and Master's Degrees ***Sept 2018 – Dec 2023***

- **Bachelor's** GPA: 9.49/10 (**Computer software and automation**) - graduated in the top of the class.
- **Master's** GPA: 9.75/10 (**Cybersecurity and payment systems**) - graduated in the top of the class.
- Recipient of the **National Fellowship**, a competitive award recognizing top 1% of the students.
- Placed 5th / 60 teams in the **National CyberHero hackathon (CTF),** solving security challenges across multiple domains.